

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (ISE) (Credit System) Degree Examinations

Supplementary Examination - May - June 2023

20IS503 - FINITE AUTOMATA AND FORMAL LANGUAGES

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**

Unit – I

Marks BT* CO* PO*

1. a) Illustrate with an example the application of automata theory to digital design.
- b) Obtain a DFA to accept strings of a's and b's having
 - i) even number of a's and b's.
 - ii) length of string is divisible by 5
- c) Convert the following NFA to it's equivalent DFA

	0	1
p	{p,q}	{p}
q	{r}	{r}
r	{s}	ϕ
*s	{s}	{s}

2. a) Define distinguishable states, indistinguishable and inaccessible states.
- b) Obtain the regular expression for the following DFA.

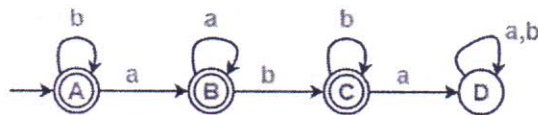


Fig. 2.b)

- c) Find a grammar that generates
 - i) $L = \{ w : n_a(w) = n_b(w), w \in \{a,b\}^* \}$
 - ii) $L = \{ wcw^R : \text{where } w \in \{a,b\}^* \}$.
3. a) Construct the minimum state equivalent DFA.

States / Σ	0	1
-->A	B	C
B	C	E
*C	D	C
D	C	E
*E	B	E

- b) Define regular expression. Find the regular expression for the language over $\Sigma = \{a,b\}$.
 - i) $L = \{w : |w| \bmod 3 = 0\}$
 - ii) $L = \{w : w \text{ ends with } ab \text{ or } bba\}$
- c) Design a DFA to accept decimal numbers divisible by 3.

P.T.O.

Unit – II

4. a) Construct a CFG for the following languages.
 (a) $L = \{a^n ww^R b^n : w \in \{a,b\}^*\}$ (b) $L = \{a^i b^j c^k, i=j+k\}$, $\Sigma = \{a,b,c\}$ 06 L3 3 2
 b) Consider the following grammar. Generate the LMD, RMD & Parse tree for the following string.
 $W = \text{badbabaadb}$
 $S \rightarrow AaAb \mid BbBa$ $A \rightarrow aAb \mid bAB \mid d$ $B \rightarrow aB \mid bBa \mid \epsilon$ 06 L3 3 2
 c) Show that the following grammars are ambiguous.
 i) $E \rightarrow E+E \mid E * E \mid (E) \mid id$
 ii) $S \rightarrow SS \mid aSb \mid bSa \mid \epsilon$ 08 L3 3 2
5. a) What do you mean by null production and unit production? Give an example. 06 L2 3 2
 b) Find a grammar in Chomsky Normal form equivalent to
 $S \rightarrow aAD$; $A \rightarrow aB \mid AB$; $B \rightarrow b$, $D \rightarrow d$. 06 L3 3 2
 c) Design a PDA for accepting a language $\{a^n b^{2n}, n \geq 1\}$. 08 L3 4 3
6. a) Construct a PDA accepting $\{a^n b^m a^n : m, n \geq 1\}$ by empty stack. 10 L3 4 2
 b) Find the language generated by CFG $G = (\{S\}, \{a,b\}, \{S \rightarrow aSb, S \rightarrow ab\}, S)$ 04 L3 3 1
 c) Convert to Greibach Normal form the grammar
 $S \rightarrow ABb \mid a$, $A \rightarrow abA \mid b$, $B \rightarrow aAb$ 06 L3 3 1

Unit – III

7. a) Prove that the regular languages are closed under concatenation and complement. 06 L2 5 2
 b) Write a DFA to accept the intersection of $L_1 = (a+b)^*a$ and $L_2 = (a+b)^*b$. 06 L3 5 2
 c) Design a Turing machine to accept the language,
 $L = \{w : w \in \{0,1\}^* \text{ and end with } 011\}$ 08 L4 5 3
8. a) Show that $L = \{a^n b^n c^n \mid n \geq 0\}$ is not Context free. 06 L3 5 2
 b) Define the right quotient of L_1 with L_2 .
 If $L_1 = \{a^n b^m \mid n \geq 1, m \geq 0\} \cup \{ba\}$ and $L_2 = \{b^m \mid m \geq 1\}$, find the right quotient of L_1 with L_2 . 08 L3 5 2
 c) Design a Turing machine to compute $x + y$, for x and y positive integers represented in unary. 06 L4 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
